Nguyen Hoang Anh

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AI Engineer Intern / Fresher

Computer Science student focused on NLP and applied machine learning. Project experience includes semantic similarity pipelines, neural machine translation, and reproducible time-series experiments using PyTorch.

EDUCATION

Sai Gon University (SGU), Ho Chi Minh City | Expected June 2027
B.Sc. in Computer Science |

Relevant coursework: Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Database Systems.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, HTML/CSS
ML & NLP: PyTorch, Hugging Face Transformers, Sentence-Transformers, scikit-learn, scikit-surprise, spaCy, TextBlob
Data & Visualization: NumPy, pandas, Matplotlib, Seaborn; EDA, data cleaning, feature engineering, cross-validation
Methods: Machine Learning, Deep Learning, NLP, Recommendation Systems
Frontend: React, React Router, Zustand, Axios, Vite
Backend: ASP.NET Core, Entity Framework Core, REST APIs, SignalR, JWT authentication, BCrypt
Data Collection: Web scraping
Engineering & Databases: Git, Docker, Jupyter, MySQL, SQLite

SELECTED PROJECTS

CaTE-X | AI Paraphrase Detection

github.com/hoagannhh/Specialized-Project

Technologies: Python, PyTorch, Transformers, Sentence-Transformers, scikit-learn

- Implemented a five-stage pipeline combining TF-IDF, pretrained sentence embeddings, a triplet encoder, a cross-encoder, and LLM-generated explanations; added validation-based score fusion.
- Built Humarin and PADBen data preparation with shared source-group splits, duplicate and label-conflict checks, and random / lexical-hard negatives sampled within each split.
- Developed training and evaluation workflows with checkpoint resume, validation-selected thresholds, document source retrieval, and highlighted HTML reports.

Hybrid Models for Stock Price Forecasting

github.com/hoagannhh/Hybrid-Computational-Intelligence

Technologies: Python, PyTorch, pandas, scikit-learn, Wavelet transforms, PSO

- Implemented an experiment pipeline for five model configurations: LSTM, LSTM-PSO, Wavelet-LSTM, Wavelet-LSTM-PSO, and ANFIS, alongside a naive evaluation baseline.
- Used chronological 70/15/15 splits, training-only scaling, and past-context windows; restricted PSO hyperparameter search to training and validation data.
- Added configurable multi-seed runs, early stopping, checkpoint loading, learning curves, and per-seed / aggregate evaluation reports.

English-to-French Neural Machine Translation

github.com/hoagannhh/machine-translation-NLP

Technologies: Python, PyTorch, spaCy, NLTK

- Built a modular Seq2Seq LSTM encoder-decoder with tokenization, vocabulary construction, padded batches, and packed input sequences.
- Implemented teacher forcing, gradient clipping, early stopping, and best-checkpoint saving; added greedy and beam-search decoding with BLEU evaluation.